

Exploratory data analysis of the King County housing

Finding the right home for Nicole Johnson

Client & Objective

Nicole Johnson – Buyer

What is she looking for?

1. Lively neighborhood
2. Central location
3. Middle price range
4. Right timing within one year

Main Objective

Find the best balance between location, affordability and timing

Three hypotheses guide the analysis

1 – Location

- Houses located closer to Central Seattle tend to be more expensive than houses farther away from the city center.
- **H2 – Timing**
- House prices vary depending on the time of year, meaning that some months may offer better buying opportunities than others.
- **H3 – Size vs. Affordability**
- Smaller houses in central locations are more likely to fall within the middle price range than larger houses in the same area.

Dataset & Scope

King County housing sales · May 2014–May 2015

21,597

House sales

**May 2014–May
2015**

Observation period

King County

Seattle-area market

- Variables used: sale price, sale date, living space, ZIP code, latitude and longitude.
- The presentation focuses only on variables relevant to Nicole's decision.
- This is an exploratory analysis: observed patterns support recommendations but do not prove causality.

Assumptions & Analytical Definitions

Definitions for client requirements that are not directly measured.

\$322k–\$645k

Middle price range · 25th–75th
percentile

≤ 15 km

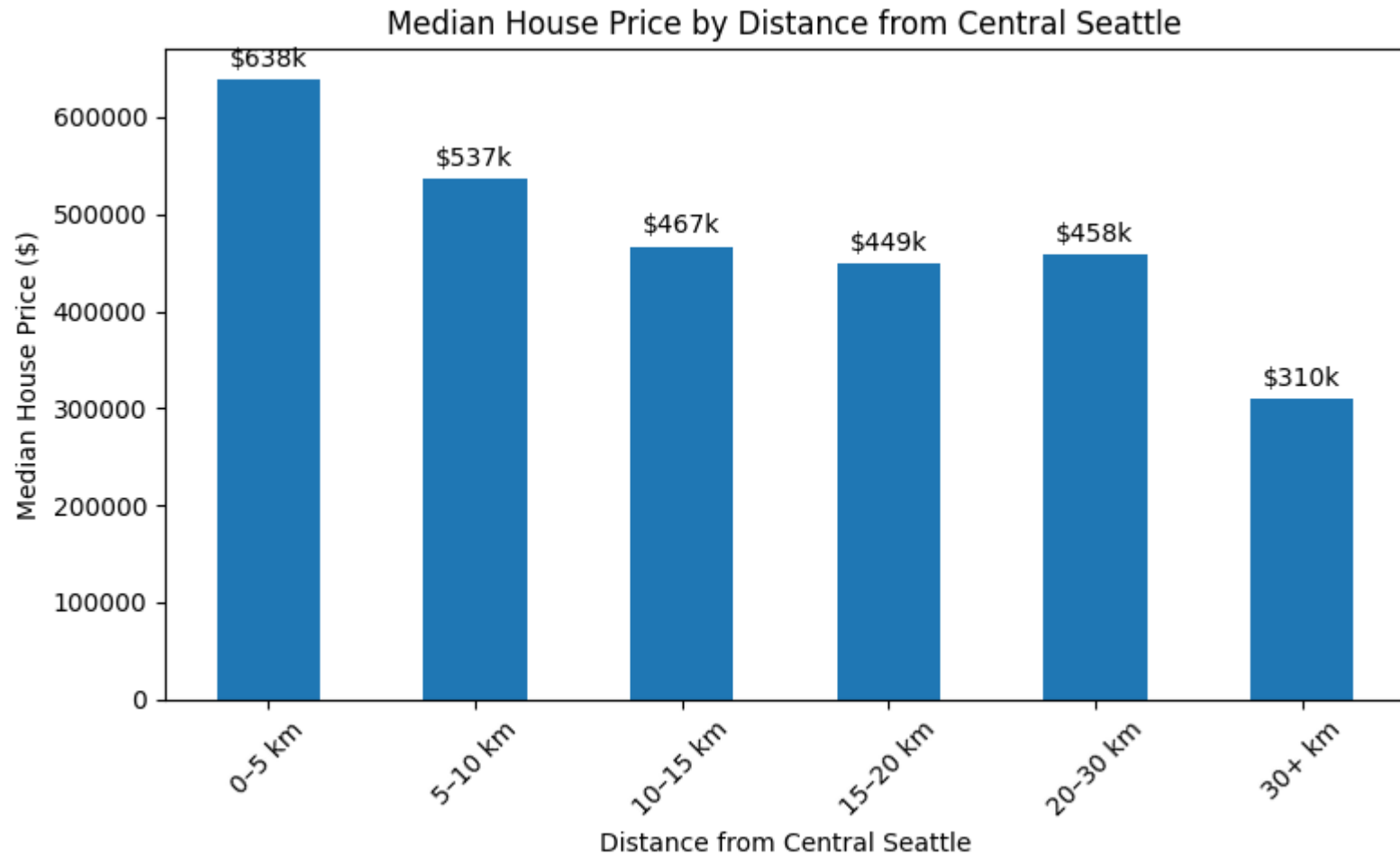
Relatively central · client analysis

Central Seattle

Reference point for distance

- “Lively neighborhood” is not directly measured in the dataset and is therefore not claimed as a proven result.
- Timing findings are based on 13 months of observed sales.
- The 15 km threshold is an analytical definition used to focus on homes that remain relatively close to Central Seattle.

H1 – House Prices by Distance from Central Seattle



H1 – What Does This Mean for Nicole?

The 5–15 km range offers a promising balance between centrality and affordability.

\$537k

Median price · 5–10
km

\$467k

Median price · 10–15 km

\$322k–\$645k

Defined middle price range

House prices generally decrease as distance from Central Seattle increases.

The 5–15 km area remains relatively central while median prices fall inside Nicole's middle price range.

Location matters, but the weak-to-moderate correlation shows that other property characteristics also influence price.

H2 – Timing for Nicole's Relevant Segment

Filtered to properties ≤ 15 km from Central Seattle and priced between \$322k and \$645k.

4,791

Relevant properties

\$453k

Lowest median ·
November

569

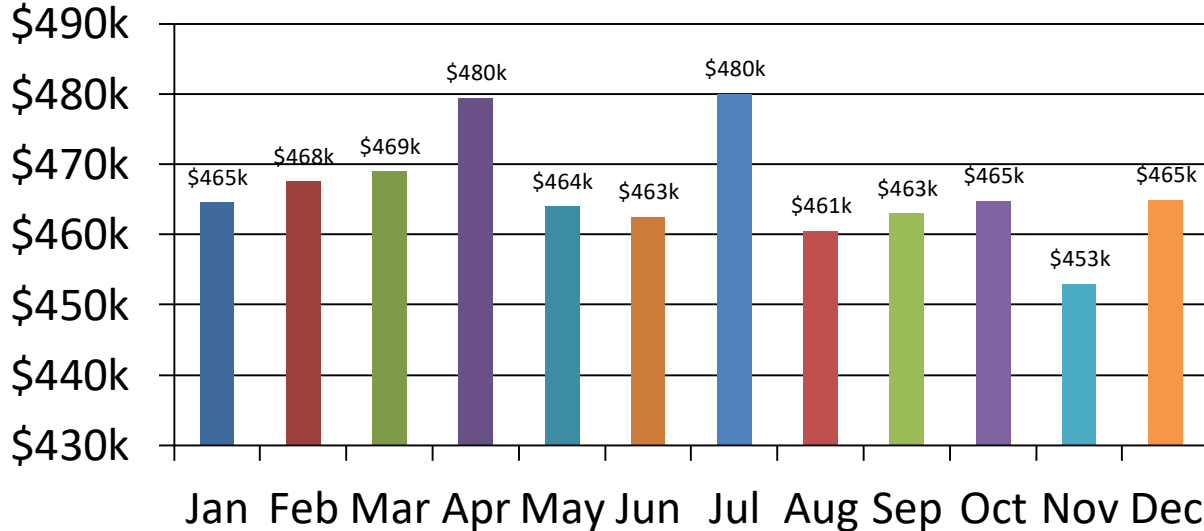
Largest selection · May

≤ 15 km

Centrality filter

Price perspective: November had the lowest observed median price.

Median price



Key Insights

The three findings that matter most for Nicole.

5–15 km

Promising location range

\$453k / 569

November price / May choice

**1,501–2,000
sqft**

Strongest size-price balance

Location: median prices generally decrease with distance from Central Seattle.

Timing: the lowest observed median and the largest selection occur in different months.

Size: larger central homes become much less likely to fit the target price range.

Recommendations for Nicole

Each recommendation is directly connected to an observed finding.

1. LOCATION → Focus primarily within 5–15 km of Central Seattle to balance centrality and affordability.
2. TIMING → If price is the priority, pay particular attention to late-year opportunities; if choice is the priority, widen the search in spring.
3. SIZE → Prioritize homes around 1,500–2,000 sqft; 67.1% of this central segment fell within the target price range.

Best strategy: optimize location, timing and living space together — not one factor in isolation.